

The Alpha Factory

A strategy factory and a four-layer validation gate · Clarion Capital take-home

Pre-registered criteria SHA-256 8e35cb...d5d8eab · full technical report in `report/report.pdf`

Verdict: 48 candidates in, 0 survivors out. A factory builds 48 strategies from two opposed market hypotheses; a four-layer gate, run independently on all of them, lets none through. The strongest, an ETH trend at 1.17 net Sharpe, fails on two counts: it breaches the drawdown cap in the 2020 crash, and its Deflated Sharpe is 0.35 against a 0.95 bar. The deliverable is the machine and the honest result, not an alpha.

Strategies	Opposed families	Markets	Survivors	Best net Sharpe	Best Deflated Sharpe
48	2	4	0	1.17	0.35 (0.95 to pass)

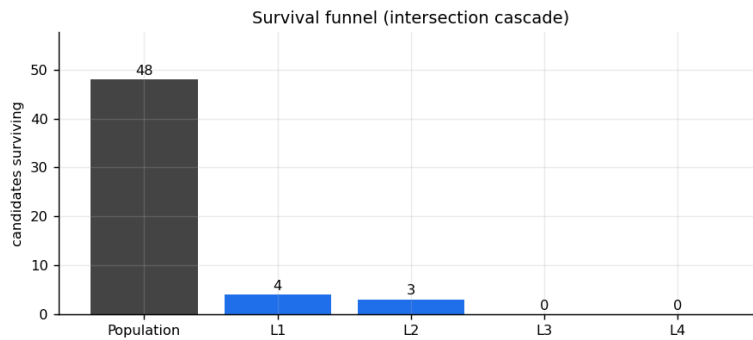


Figure 1: The survival funnel: 48 candidates enter, 0 pass all four independent layers.

The factory. 48 instances = 2 opposed families $\times$ 6 parameter settings $\times$ 4 markets (S&P 500, USD/JPY, gold, ETH), each with at most two free parameters. Being opposed by construction, their signals are negatively correlated (≈ -0.30) and their returns nearly uncorrelated across families: genuine diversification, not one bet in disguise.

Family	The bet	Free parameters
voltarget_mom	Trend: ride the move, sized to constant vol (cap $2\times$)	lookback, vol window
bollinger_revert	Reversal: fade extremes, only in calm regimes	lookback, z -threshold

How the gate works

Four independent, pre-registered tests run on the full population; survivors are their *intersection*, not a sequential filter, which is what exposes that L4 alone rejects all 48.

Layer	What it tests	Pass	Dominant failure
L1	Out-of-sample edge + $\pm 20\%$ robustness	4 / 48	OoS Sharpe below bar
L2	Walk-forward stability across windows	6 / 48	out-of-window Sharpe/return low
L3	Stress: COVID + own worst windows + $2\times$ vol	8 / 48	blows up in the COVID window
L4	Block-permutation significance + Deflated Sharpe	0 / 48	fails the Deflated Sharpe
Survivors = intersection of all four		0 / 48	

Why nothing survives

The binding constraint is the **Deflated Sharpe**, which corrects a Sharpe for the number of strategies tried. The luckiest of 48 zero-skill trials would be expected to score ≈ 1.32 annualised; the best candidate's 1.17 sits *below* that luck hurdle, for a Deflated Sharpe of just 0.35. Crediting the correlation among trials (an effective ≈ 12 rather than 48) lifts it only to 0.69, still short of 0.95. Generating *more* candidates cannot manufacture a survivor, because each one raises the bar for all of them.

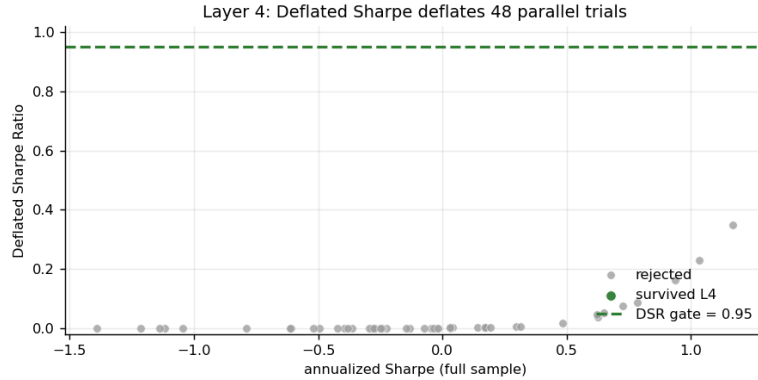


Figure 2: Deflated Sharpe vs. net Sharpe across the 48. None clears 0.95; the best reaches 0.35.

Where the costs bite

The two families fail for different reasons: reversion is low-turnover but weak *even gross*, so costs are not what kill it; trend is high-turnover (it re-scales every bar to hold volatility constant) and is where costs bite, the median trend instance falling from a gross Sharpe of 0.59 to 0.11 net.

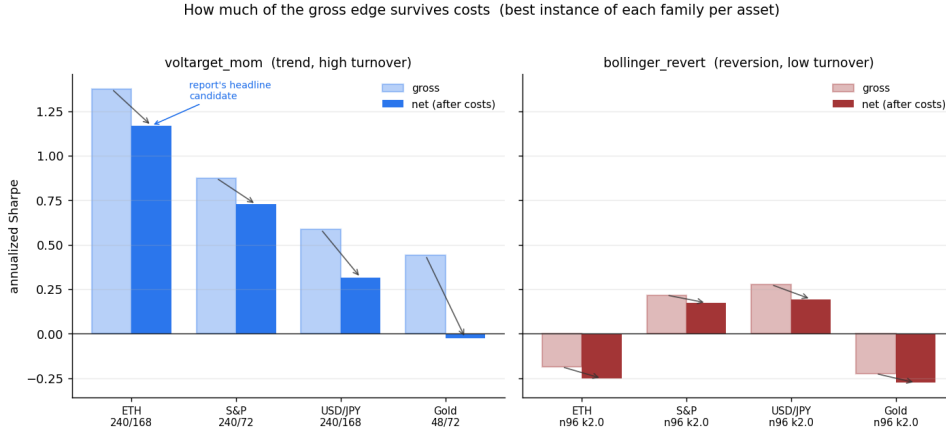


Figure 3: Gross vs. net Sharpe, best instance per market. Costs barely touch the low-turnover reversion grid but materially erode the high-turnover trend grid.

Honesty, reproducibility, and what comes next

Every threshold was fixed in `config/gate_criteria.yaml` and SHA-256-hashed *before* the gate ran; the hash held across every strategy iteration (one documented L2 redesign aside), so no bar was ever tuned to admit a winner. The pipeline is deterministic and reproduces $48 \rightarrow 0$ from the raw data with `python run.py`. And the path forward is *not* a richer factory, since more variations only raise the deflation bar: it is more history, genuinely orthogonal alpha (new data or a different economic rationale), and combining several weak but uncorrelated signals at the portfolio level. Better ideas and more data, never a looser gate.